



Optical Fibres

Operating Instructions

1 Warnings

1. Always wear protective glasses or goggles when working with glass fibres. Small slivers of glass are hard to pick out of eyes.
2. Take care not to poke glass fibres into your skin. Collect broken pieces of fibres on a piece of sticky tape and dispose of in the yellow "sharps waste" provided.
3. Do not eat or drink during the experiment.
4. Do not look into fibres when using a laser as the light source.
5. The light source used in these experiments is a Class 2 He-Ne laser which is capable of damaging the retina of the eye if the laser beam (or a specular reflection) is viewed directly.
6. Keep the working bench clear of unnecessary apparatus to minimise the chance of strong stray reflections and make sure the main beam and any strong reflections are terminated on your experimental bench (e.g. by using a screen).

2 Experiment 1: Preparation of Fibre Ends

2.1 Experimental Equipment



Jacket Stripper



CT-30 Fiber Cleaver (locked)

2.2 Experimental Instructions

1. Remove about 1-2 cm of jacket from each end of a 1-1.5 m segment of the F-MLD multimode fibre (please use the pre-cut lengths of fibre rather than shortening the 1500 m spool).
 - The Newport fibre jacket stripper is well suited for F-MLD fibres. Opposing spring loaded jaws of the stripper grip the fibre over a large area while axial tension is applied to remove the coating.
 - The F-MLD fibre has an outer green layer (the coating), a middle clear layer (the cladding) and an inner glass layer (the core). You must remove both the coating and the cladding to cleave the fibre.
2. Clean the fibre in acetone-soaked tissue.
3. Remove about 1-2 cm of jacket from each end of a 1.5-2 m segment of the F-SV single-mode fibre; by dipping the fibre end into acetone for about 5 minutes and stripping the jacket with a tissue soaked with acetone. Alternatively you could try to use the Newport fibre jacket stripper again and clean the fibre with the acetone-soaked tissue.
4. Use the CT-30 Fibre Cleaver to cleave the fibre:

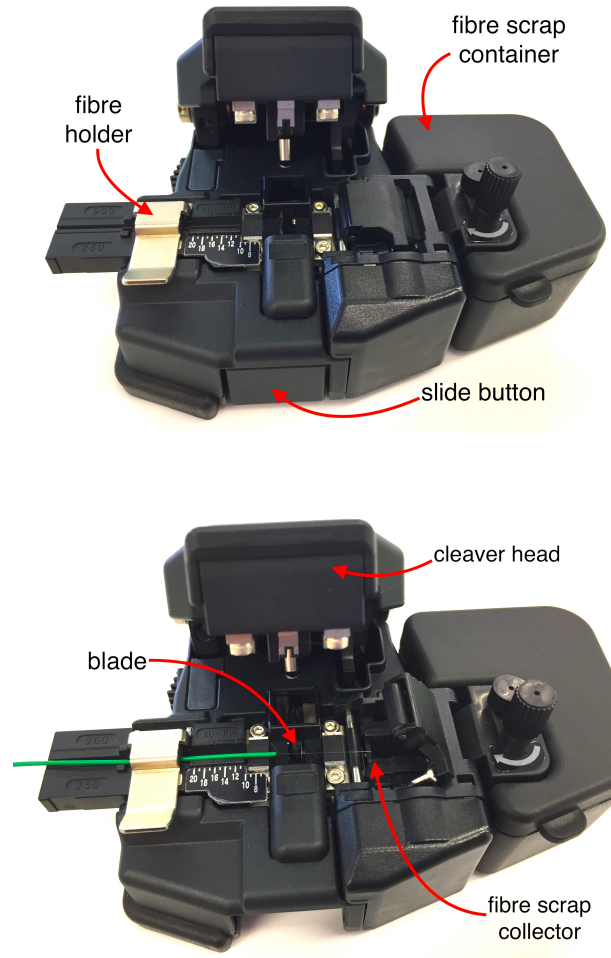


Figure 2: CT-30 Fibre Cleaver with and without slide button engaged and fibre in place.

Cleaving with CT-30:

- Strip 24 to 25 mm of coating from the fibre end.
 - Clean the exposed fibre cladding.
 - Push the *slide button* until it locks.
 - Set the stripped optical fibre on the cleaver.
 - Press down on the *cleaver head*. A spring force will bring it to its open position.
 - The *fibre scrap collector* rollers drive the fibre scrap into the *fibre scrap container* automatically.
 - **Do not put a finger in the driving area of the slide button as personal injury may result.**
5. Check the quality of the cleave side-on with a microscope. The end face should appear **flat** and **free of defects**. However, small chips or cracks which do not extend into the central region of the fibre are acceptable. Well cleaved and poorly cleaved fibres are compared in Figure 1:

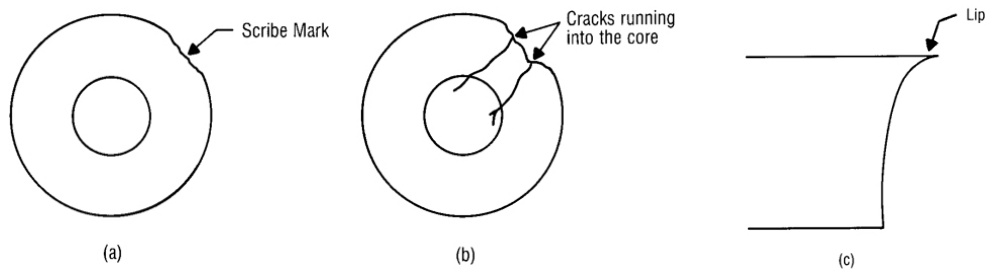


Figure 3: Cleaved fibre ends. (a) Good cleave. (b) Cracked fibre. (c) Side view of lip.

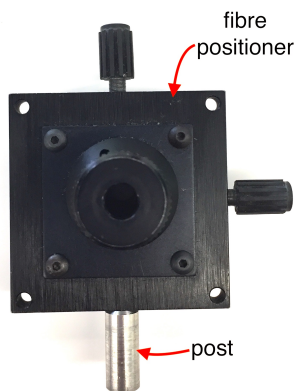
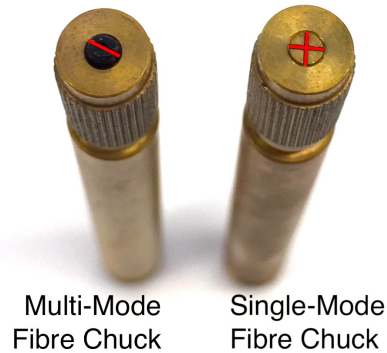
6. Launch light from a lamp into the cleaved end of F-MLD fibre and examine the geometry of the fibre.

3 Experiment 2: Measurement of Numerical Aperture

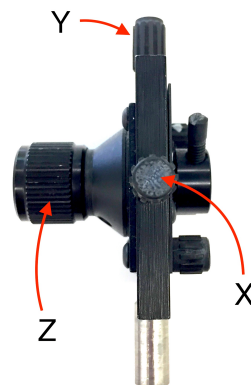
3.1 Experimental Equipment



Multi-mode and single-mode fibre chucks



Fibre positioner and post



Side view of fibre positioner with x-y-z adjustment screws



Power meter



20X microscope objective

3.2 Experimental Instructions

a) Numerical Aperture Using Incident Angle and Power Accepted by Fibre

1. Use the He-Ne laser marked for multimode fibres (furthest from the bench). This laser should be equipped with a 20X microscope objective, while the laser for single-mode fibres requires a 10X microscope objective.
2. Mount the rotation stage so that the He-Ne laser beam passes over the centre hole of the rotation stage. The beam should fall in the centre of a small white spot marked on the screen. Mount the small post holder (MPH-1) on the rotation stage.
3. Insert a cleaved end of the F-MLD multimode fibre into the fibre chuck of a fibre positioner. Mount the fibre positioner on a short post and insert the post into the post holder on the rotation stage. **NB:** The cleaved fibre can be **easily damaged** during insertion into the fibre chuck. Threading the fibre through the chuck before cleaving or re-cleaving the fibre can help.
4. Extend the tip of the fibre and orient the fibre positioner so that the fibre tip is at the centre of the rotation stage. Make sure that the tip of the fibre remains at the centre of the laser beam as the stage is rotated. The alignment is critical to obtain an accurate value for the fibre NA (numerical aperture).
5. Mount the far end of the fibre in the chuck of a fibre positioner. Mount the fibre positioner onto a post. Mount the detector head of the power meter so that the output beam of the fibre is incident on the detector. You may need a hood of black paper to prevent stray room light entering the detector. Block the laser beam and note down the power level of the stray light.
6. Measure the power accepted by the fibre as a function of the incident angle of the plane-wave laser beam. Use both positive and negative rotation directions to compensate for any laser-fibre misalignment. Rotation angles between $+30^\circ$ and -30° in steps of 2° are sufficient. Correct your data for the background stray light.
7. Make a semi-log plot the power accepted by the fibre as a function of the sine of the incident angle. The experimental value for the numerical aperture of the fibre is the half-width of the curve at the points where the accepted power is 5% of the maximum value.

b) Numerical Aperture Using Far-Field Distribution

1. Remove the detector head and place a card at a distance of about 8cm, L , from the fibre end to display the laser spot coming out of the fibre.
2. Measure the width, W , of the laser spot.
3. The approximate measure of NA is $W/2L$. Compare this value with that found in the first half of this experiment and the value specified by the manufacturer.

4 Experiment 3: Optical Fibre Attenuation

4.1 New Experimental Equipment



F-MLD 1500 m fibre spool

4.2 Experimental Instructions

1. Prepare both ends of the 1500 m fibre spool. Note down the actual length recorded on the spool. This fibre is Newport F-MLD multimode fibre with a $100\ \mu\text{m}$ core.
2. Mount a fibre positioner onto a post. Place the cleaved far end of the fibre into a chuck and insert the chuck into the positioner. Align the detector head of the power meter with the fibre end.
3. Mount the F-916 fibre coupler near the He-Ne laser. Use a 20X objective lens. Align the laser and coupler so that laser beam is parallel to the lens axis and strikes the centre of the lens. Place the cleaved input end of the fibre into the fibre chuck and insert this into the coupler.

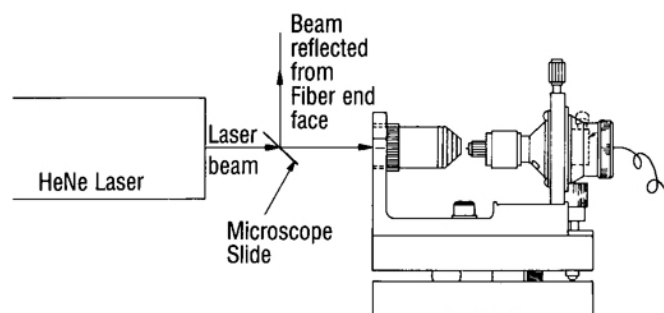


Figure 8: Coupling of He-Ne laser light into a fibre using F-916 fibre coupler.

4. Use the x, y and z position adjustments on the positioner to locate the end of the fibre in the focused beam. You may insert a microscope slide temporarily in the path of the laser beam to look at the Fresnel reflection from the fibre end face and focus the Fresnel reflected beam by adjusting the z-position of the fibre.

5. Adjust the x and y position knobs on the fibre positioner until some light is emitted from the output end of the fibre. Then optimise the output with the fine position controls on the tile stage while monitoring the power output from the far end.
6. Study the FM-1 mode scrambler instructions that follow this experiment. Position the mode scrambler at a convenient place near the launch end (i.e. input end) of the fibre.
7. Rotate the knob of the mode scrambler counter-clockwise to fully separate the two corrugated jaws. Place the fibre between them. Rotate the knob clockwise until the jaws just touch the fibre. You should be able to see a faint glow in the fibre at the position where the jaws are grabbing. Note down the power output before mode scrambling.
8. Note down the position of the knob on its graduated scale. Rotate the knob further clockwise for 3-7 major divisions to attain a stable mode distribution. Note down the power output again. Comment on the difference, if any. **This launching and mode scrambling set-up should be maintained until the end of the next step.**
9. Cut the fibre about 2 m after the mode scrambler. Cleave the broken end and measure the output power from the cutback segment. Calculate the attenuation coefficient. Compare this with the specified value. Explain any difference.
10. Release the fibre from the corrugated jaws of the mode scrambler by turning the knob counter-clockwise. Note down the power output. Discuss the difference, if any.

5 FM-1 Mode Scrambler

Newport's FM-1 Mode Scrambler uses a precision translation stage to press a short length of multi-mode fibre between specially designed surfaces that introduce micro-bending in the fibre. This dramatically increases coupling among the guided and radiation modes of the fibre. (The FM-1 is begin used as a mode scrambler when low-order modes have been preferentially excited. It is being used as a mode filter when mostly high-order modes have been launched). The FM-1 repeatably achieves a stable modal distribution in the fibre without requiring kilometres of intervening fibre.

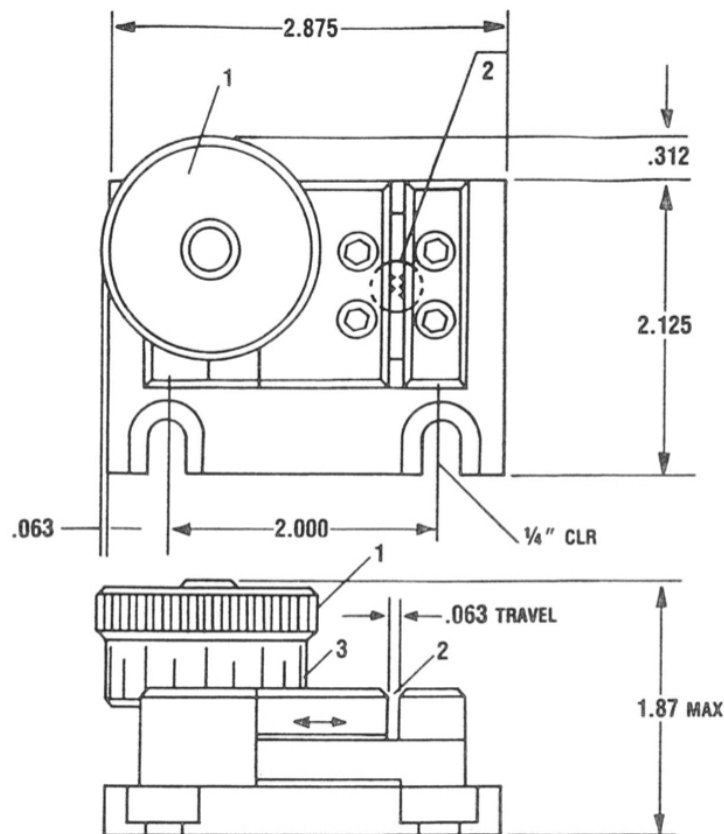
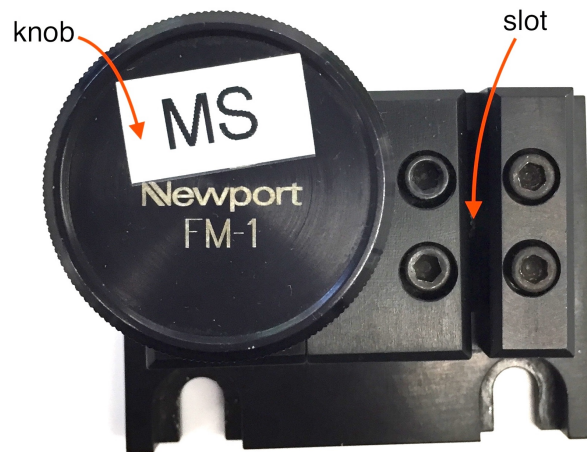


Figure 9: FM-1 Mode Scrambler Top and Side View Schematics

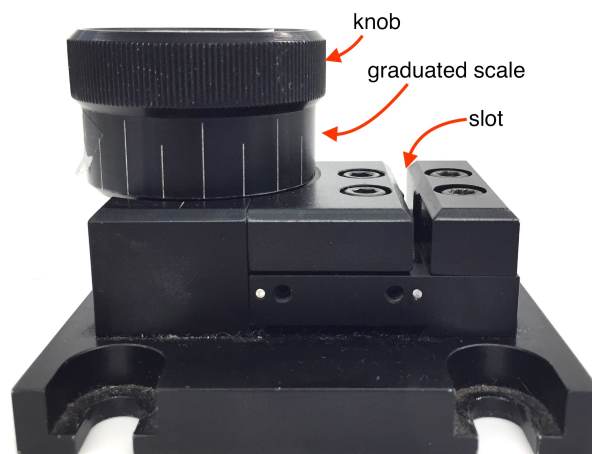
5.1 Using the FM-1

1. Place the FM-1 near the input end of the fibre (See Fig)
2. Rotate the knob, (1), counter-clockwise to fully separate the two corrugated surfaces.
3. Place the fibre between the corrugated surfaces and into the slot, (2). No special preparation of the fibre is required; it is suggested that the fibre jacket be left on to protect the fragile glass fibre.

4. Rotate the knob clockwise until the corrugated surfaces just contact the fibre. Note the position of the knob on its graduated scale, (3). Each major graduation represents a $25\text{ }\mu\text{m}$ change in the separation of the corrugated surfaces; the smaller graduations represent $12.5\text{ }\mu\text{m}$ displacements.
5. Most laboratory applications require 75 to $175\text{ }\mu\text{m}$ of displacement (three to seven major graduations) for adequate micro-bending and mode filtering. (Some soft fibre jackets will require a brief stabilisation period after the micro-bending displacement is set.)



FM-1 Mode Scrambler Top View



FM-1 Mode Scrambler Side View

6 Experiment 4: Single-Mode Fibres

6.1 New Experimental Equipment



10X microscope objective

6.2 Experimental Instructions

1. Follow steps 3-6 of experiment 3 to couple He-Ne laser light into the F-SV single-mode fibre segment that you have already prepared in experiment 1 (using the single-mode fibre chucks and 10X objective lens). A distance of about 50 cm between the laser and the microscope objective will be sufficient to achieve a reasonably good coupling efficiency. A coupling loss of 3 dB or less is probably achievable.
2. Mount the far end of the fibre on the rotation stage in exactly the same way as the front end of the fibre was mounted in experiment 2. The end of the fibre should be on the axis of the rotation stage.
3. Mask the detector head with black card-board which has a narrow slit in the middle. Mount it on the table on a post about 10 cm from the fibre end face.
4. Rotate the tip of the fibre by turning the rotation stage. This will effectively scan the far-field distribution of the fibre across the detector. Measure the power received by the detector as a function of the angular position of the fibre. An angular scan from -8° to $+8^\circ$ with 1° steps will be sufficient.
5. Plot the far-field distribution of the fibre as a function of angle.
 - (a) Note down the maximum recorded power from your data as $I(0)$.
 - (b) Measure the full width between the points where the output power is $I(0)e^{-2}$.
 - (c) Take half of that full width as θ_A .
 - (d) Calculate $\sin \theta_A$ and compare your result with the fibre's numerical aperture quoted by the manufacturer.

6. On the same graph, plot a Gaussian curve:

$$I(\theta) = I(0)e^{-(\frac{\theta}{\theta_0})^2}$$

Where $\theta_0 = \frac{\theta_A}{\sqrt{2}}$.

Assess the validity of the Gaussian approximation.